

An Introduction to Topic Modelling in NLP

Topic modelling in natural language processing is used to categorise information, organise huge text data, obtain a summary of a large corpus, and improve recommendation systems by identifying commonalities within the corpus. Let's explore the LDA technique to implement the topic modelling of a corpus.



In natural language processing (NLP), topic modelling is an automatic unsupervised machine learning technique that determines the abstracts (topics) present in a large corpus. Words with analogous meanings are grouped into topics, which helps in finding patterns in a textual data corpus without a training dataset.

Topic modelling techniques

Among the common topic modelling techniques, Latent Dirichlet Allocation (LDA), Non-Negative Matrix Factorization (NMF) and Latent Semantic Analysis (LSA) are used the most. Among these three, LDA is the most efficient and popular. It is a generative probabilistic model and considers that topic words are distributed over the documents. It uses Dirichlet priors to ensure that topic words dominate the documents. We shall now discuss the LDA technique to implement the topic modelling of a corpus.

To implement topic modelling 'choosing the right number of topics' is essential, since too few topics oversimplify the task and too many lead to an overlap. Handling 'stopwords' and 'noisy data' needs special care to ensure meaningful outcomes of topics. Most importantly, human interpretation is needed to identify the proper topic from the cluster of output words.

Topic modelling can be used for high-level summarisation of a text corpus and for:

- *Customer feedback analysis:* Identifying recurring themes

in customer reviews.

- *Fake news detection:* Analysing and classifying news articles.
- *Medical research:* Discovering patterns in clinical literature.
- *Legal document analysis:* Categorising case laws and contracts.

LDA components

LDA is often used for topic identification in large text corpora. It can be used to group similar documents based on their shared topic. In a literature survey, this technique is used to identify similar articles. Text summarisation is another important application of this technique. LDA consists of three key components.

Documents as topic mixtures: Each document is represented as a mixture of multiple topics, where each topic has a probability distribution. This is called alpha control, where high values indicate more topics within the documents.

Topics as word distributions: Here, each topic is characterised by a probability distribution over words and is called beta control. It controls the topic-word distribution; a high value represents topics as a mixture of more words.

Dirichlet priors: The model assumes Dirichlet distributions for the topic distribution in documents and the word distribution in topics, which helps in controlling sparsity. There is a hidden variable Z , which represents the latent topic assignments for each word in a document.

LDA algorithm

1. **Initialization:** Randomly assign the topics to each word in each document.
2. **Iterate:**
 - Based on the frequency of words, reassign topics to words.
 - Apprise the topic distributions over words and the document distributions over topics.
3. **Convergence:** Iterative step 2 continues till the topic distribution is not stable.